

Logaritmo

$$\log_a b = x$$
$$a^x = b$$

$$\log_a (b \cdot c) = \log_a b + \log_a c$$

$$\log_a (b/c) = \log_a b - \log_a c$$

$$\log_a 1 = 0$$

$$\log_a b^c = c \log_a b$$

$$\log_a a^x = x$$

$$\log_a b = \frac{\log b}{\log a}$$

$$\log_a a = 1$$

$$a^{\log_a b} = b$$

$$a^m \cdot a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a \cdot b)^n = a^n \cdot b^n$$

$$(a^n)^m = a^{n \cdot m}$$

$$a^0 = 1 \quad a^1 = a$$

$$a^{-m} = \left(\frac{1}{a}\right)^m$$

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

TRANSFORMADA DE LAPLACE

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt \quad | \quad \mathcal{L}\{f(t)\} = F(s)$$

TRANSFORMADA DE ALGUNAS FUNCIONES

$$\mathcal{L}\{1\} = \frac{1}{s}$$

$$\mathcal{L}\{\sinh kt\} = \frac{k}{s^2 - k^2}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2} \quad \mathcal{L}\{t^m\} = \frac{m!}{s^{m+1}}$$

$$\mathcal{L}\{\cosh kt\} = \frac{s}{s^2 - k^2}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

$$\mathcal{L}\{\sin(kt)\} = \frac{k}{s^2 + k^2}$$

Ejemplos $\mathcal{L}\{t^2\} = \frac{2!}{s^3}$ $\swarrow f(t)$

$$\mathcal{L}\{\cos kt\} = \frac{s}{s^2 + k^2}$$

$$\mathcal{L}\{e^{4t}\} = \frac{1}{s-4}$$

TRANSFORMADA INVERSA

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 - k^2}\right\} = \cosh kt$$

$$\mathcal{L}^{-1}\left\{\frac{m!}{s^{m+1}}\right\} = t^m$$

$$\mathcal{L}^{-1}\left\{\frac{k}{s^2 - k^2}\right\} = \sinh kt$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$$

$$\mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\} = \sin kt$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + k^2}\right\} = \cos kt$$

Transformada de Laplace

$$\mathcal{L}\{k\} = \frac{k}{s}$$

PROPIEDADES DE LINEALIDAD

$$\mathcal{L}\{a f(t) + b g(t)\} = a \mathcal{L}\{f(t)\} + b \mathcal{L}\{g(t)\}$$

$$[\mathcal{L} f(x) + \beta g(x)]' = \mathcal{L} f'(x) + \beta g'(x)$$

$$\int [\alpha f(x) + \beta g(x)] dx = \alpha \int f(x) dx + \beta \int g(x) dx$$

TRANSFORMADA DE UNA DERIVADA

$$F(s) = \mathcal{L}\{f(t)\}$$

$$\mathcal{L}\{f'\} \rightarrow \mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f\} = F(s)$$

$$\mathcal{L}\{f''\} \rightarrow \mathcal{L}\{f''(t)\} = s^2 F(s) - s f(0) - f'(0)$$

$$\mathcal{L}\{f'''\} \rightarrow \mathcal{L}\{f'''(t)\} = s^3 F(s) - s^2 f(0) - s f'(0) - f''(0)$$

PROPIEDAD DE LA TRASLACIÓN

$$\mathcal{L}\{e^{at} f(t)\} = \mathcal{L}\{f(t)\}(s-a)$$

to deriv:

$$\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}$$

$$\mathcal{L}\{e^{at} \cos(bt)\} = \frac{(s-a)}{(s-a)^2 + b^2}$$

$$\mathcal{L}\{e^{at} \sin(bt)\} = \frac{b}{(s-a)^2 + b^2}$$

(Ejemplo)

$$\mathcal{L}\{t^3 e^t\} = \frac{3!}{(s-1)^4}$$

$$(s-1)^4$$

$$(a=1)$$

$$(n=3)$$

$$\mathcal{L}\{0\} = 0$$

Transformada de Laplace

$$\mathcal{L}\{1\} = \frac{1}{s}$$

$$\mathcal{L}\{e^{kt}\} = \frac{1}{s-k}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{t^n e^{kt}\} = \frac{n!}{(s-k)^{n+1}}$$

$$\mathcal{L}\{\sin kt\} = \frac{k}{s^2+k^2}$$

$$\mathcal{L}\{\cos kt\} = \frac{s}{s^2+k^2}$$

$$\mathcal{L}\{\sinh kt\} = \frac{k}{s^2-k^2}$$

$$\mathcal{L}\{\cosh kt\} = \frac{s}{s^2-k^2}$$

$$\mathcal{L}\{\delta(t-k)\} = e^{-ks}$$

Transformada de Laplace inversa

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s-k}\right\} = e^{kt}$$

$$\mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$$

$$\mathcal{L}^{-1}\left\{\frac{n!}{(s-k)^{n+1}}\right\} = t^n e^{kt}$$

$$\mathcal{L}^{-1}\left\{\frac{k}{s^2+k^2}\right\} = \sin kt$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+k^2}\right\} = \cos kt$$

$$\mathcal{L}^{-1}\left\{\frac{k}{s^2-k^2}\right\} = \sinh kt$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2-k^2}\right\} = \cosh kt$$

$$\mathcal{L}^{-1}\{e^{-ks}\} = \delta(t-k)$$

m N° de términos
de la serie
m = 1, 2, 3, ...

SERIES DE FOURIER

$$d = 2p$$

$$\frac{d}{2} = p$$

DEFINIDA EN EL INTERVALO $(-p, p)$; $m = 1, 2, 3, \dots$

$$f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty} \left[a_m \cos\left(\frac{m\pi}{p} x\right) + b_m \operatorname{sen}\left(\frac{m\pi}{p} x\right) \right]$$

Coeficientes
de
FOURIER

$$a_0 = \frac{1}{p} \int_{-p}^p f(x) dx$$

$$a_m = \frac{1}{p} \int_{-p}^p f(x) \cos\left(\frac{m\pi}{p} x\right) dx$$

$$b_m = \frac{1}{p} \int_{-p}^p f(x) \operatorname{sen}\left(\frac{m\pi}{p} x\right) dx$$

Recordar:

$$\cos(m\pi) = (-1)^m$$

$$\operatorname{sen}(m\pi) = 0$$

Números complejos $Z = (a, b)$

El Módulo $|z| = \rho = \sqrt{a^2 + b^2}$

El Argumento φ $\operatorname{tg} \frac{b}{a} = \frac{b}{a}$ $0 \leq \varphi < 2\pi$ $\cos \frac{a}{\rho} = \frac{a}{\rho}$ $\operatorname{sen} \frac{b}{\rho} = \frac{b}{\rho}$

Forma trigonométrica $Z = \rho (\cos \varphi + i \operatorname{sen} \varphi)$

Potenciación de exponente natural (fórmula de De Moivre)

$$Z^n = \rho^n (\cos(n\varphi) + i \operatorname{sen}(n\varphi))$$

Forma exponencial (fórmula de Euler) $e^{i\varphi} = \cos \varphi + i \operatorname{sen} \varphi$

Logaritmo natural $\operatorname{Ln} z = \ln \rho + i(\varphi + 2k\pi)$ con $k \in \mathbb{Z}$

Radicación $z = \sqrt[n]{z'}$ luego $z' = (a, b)$

$$W_k = \sqrt[n]{\rho} \left(\cos\left(\frac{\varphi + 2k\pi}{n}\right) + i \operatorname{sen}\left(\frac{\varphi + 2k\pi}{n}\right) \right)$$